

Machine Learning: The No-Nonsense (But Fun) Guide — Day 9

Decision Tree Classifier: The "20 Questions" Algorithm 🌳

1. Story & Intuition

Imagine you're playing **20 Questions**. You think of an animal, and I ask:

"Is it bigger than a dog?"

→ Yes

"Does it have stripes?"

→ Yes

"Is it a tiger?"

→ YES!

That's exactly how a Decision Tree works. It asks a series of questions, each one splitting the data, until it reaches a conclusion.

"Decision Trees: Because sometimes the best model is just a really good flowchart."

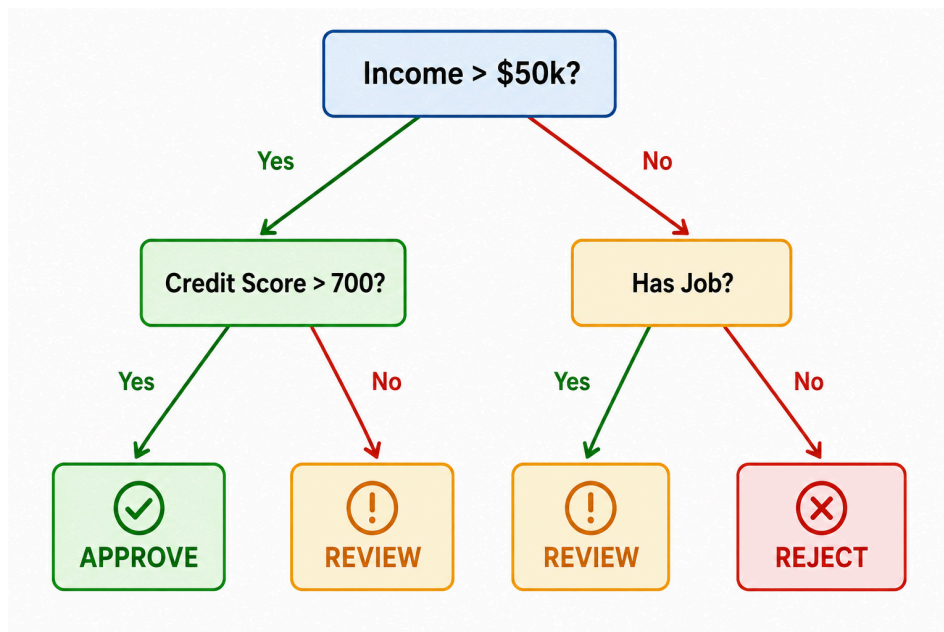
2. What Is a Decision Tree?

A Decision Tree is a **flowchart-like structure** where:

Component	What It Is	Example
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Root Node	The first question	"Is income > \$50k?"
Internal Node	A follow-up question	"Has credit card?"
Branch	The answer path	Yes / No
Leaf Node	The final prediction	"Approve loan" / "Reject loan"

Visual Representation:



3. How It Works: The Algorithm

Step 1: Pick the Best Question

How do we decide which question to ask first? We need a way to measure "how good is this split?"

Two popular methods:

Method	What It Measures	Best For
Gini Impurity	How often a random element would be mislabeled	Faster computation
Entropy / Information Gain	How much uncertainty is reduced	More theoretically sound

Gini Impurity Formula:

$$\text{Gini} = 1 - \sum(p_i)^2$$

Where p_i = probability of class i in the node.

Example:

- Node has [6 Cats, 2 Dogs]
- $\text{Gini} = 1 - (6/8)^2 - (2/8)^2 = 1 - 0.5625 - 0.0625 = \mathbf{0.375}$
- Perfect split [8 Cats, 0 Dogs]
- $\text{Gini} = 1 - (1)^2 - (0)^2 = \mathbf{0} \leftarrow \text{Pure node!}$

Entropy Formula:

$$\text{Entropy} = -\sum(p_i \times \log_2(p_i))$$

Information Gain:

$$\text{IG} = \text{Entropy}(\text{parent}) - \text{Weighted average}(\text{Entropy}(\text{children}))$$

The goal: Maximize Information Gain (or minimize Gini Impurity) at each split.

4. Building the Tree: Step by Step

Dataset:

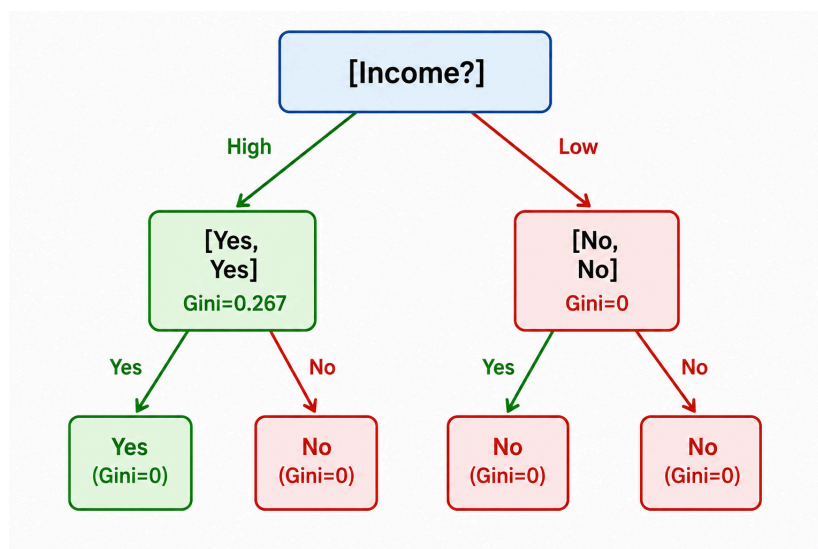
Income	Credit Score	Approved?
High	Good	Yes
High	Bad	No
Low	Good	No
Low	Bad	No
High	Good	Yes

Step 1: Calculate Gini for all possible splits.

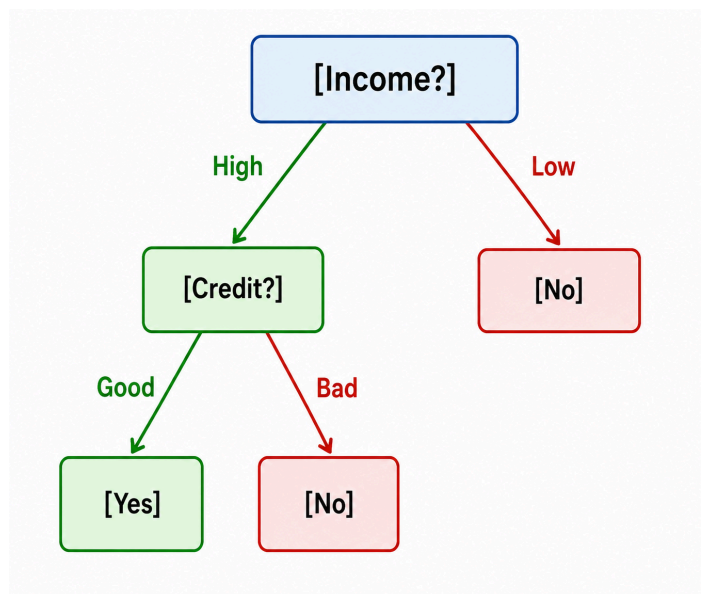
Split	Gini Impurity
Income = High/Low	0.267
Credit Score = Good/Bad	0.4

Income gives a better split → Ask Income first!

Step 2: Split on Income.



Step 3: High-income nodes still have mixed classes. Split further on Credit Score.



Step 4: All leaf nodes are pure. The tree is complete!

5. The Big Problem: Overfitting 🌳 → 🌲

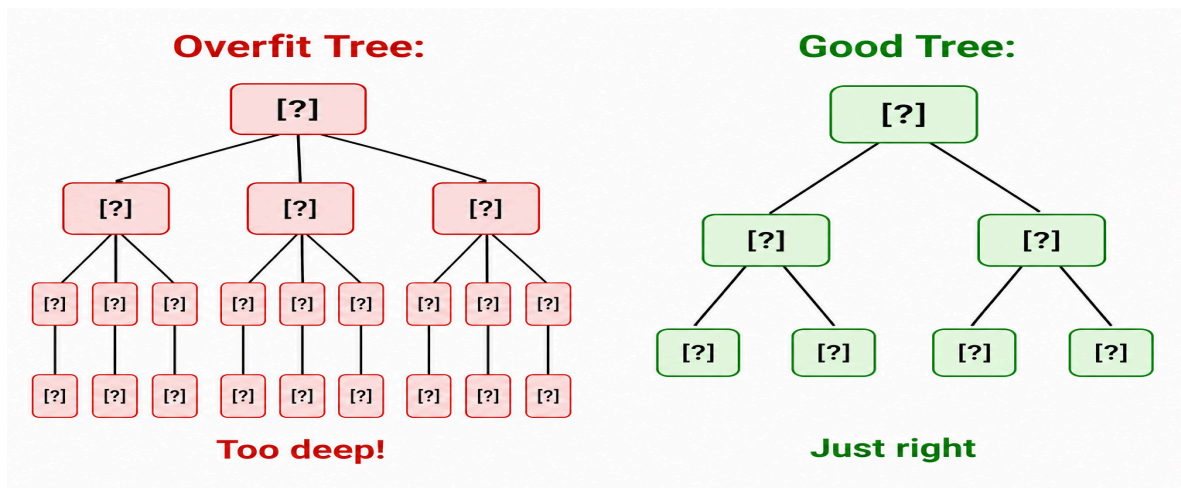
Without limits:

Training Accuracy: 100%

Testing Accuracy: 60%

The tree memorized every data point. Useless on new data.

Visual:



6. Stopping Overfitting: Hyperparameters

Hyperparameter	What It Does	Good Value
<code>max_depth</code>	Maximum tree depth	3-10
<code>min_samples_split</code>	Minimum samples to split a node	5-20
<code>min_samples_leaf</code>	Minimum samples in a leaf	2-10
<code>max_features</code>	Features to consider per split	$\sqrt{n_features}$
<code>ccp_alpha</code>	Cost-complexity pruning	0.01-0.1

In code:

```
# Pruned tree
model = DecisionTreeClassifier(
    max_depth=5,
    min_samples_split=10,
    min_samples_leaf=5,
```

```
    ccp_alpha=0.01 # pruning
)
```

7. Pruning: Cutting the Bad Branches

Pre-pruning: Stop growing early (using max_depth, min_samples, etc.)

Post-pruning: Grow a full tree, then cut back.

8. Feature Importance: What Matters Most?

Decision Trees tell you which features are most useful.

```
import pandas as pd
```

```
# Get feature importance
```

```
importance = pd.DataFrame({
    'Feature': data.feature_names,
    'Importance': model.feature_importances_
}).sort_values('Importance', ascending=False)
```

```
print(importance)
```

Output:

	Feature	Importance
2	petal length (cm)	0.513855
3	petal width (cm)	0.486145
0	sepal length (cm)	0.000000
1	sepal width (cm)	0.000000

Petal measurements matter. Sepal measurements? Not so much.

9. Decision Tree vs Other Algorithms

	Decision Tree	KNN	Logistic Regression	Random Forest
Interpretability	🏆 Excellent	Good	Medium	Good
Handles non-linear	✅ Yes	✅ Yes	❌ No	✅ Yes
Training speed	⚡ Fast	Instant	Fast	Medium
Prediction speed	⚡ Fast	🐢 Slow	⚡ Fast	Fast
Overfitting risk	High (alone)	Medium	Low	Low
Needs scaling	No	✅ Yes	Yes	No
Handles missing values	✅ Yes	No	No	✅ Yes

10. When to Use Decision Trees

✅ Use It	❌ Skip It
Need to explain decisions to humans	Need highest accuracy (use ensemble)
Mixed data types (numbers + categories)	Data has many correlated features
Feature selection needed	Very high-dimensional sparse data
Quick baseline model	Need probabilistic confidence

Small to medium datasets	Need to handle concept drift
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11. Interview Questions

1. What is Gini Impurity and how is it calculated?
 2. What is the difference between Gini and Entropy?
 3. How does a Decision Tree handle overfitting?
 4. What is pruning and why is it important?
 5. What is Information Gain?
 6. How does a Decision Tree handle categorical features?
 7. What are the advantages of Decision Trees over Neural Networks?
 8. Why is a single Decision Tree rarely used in production?
 9. What is feature importance in Decision Trees?
 10. How would you handle imbalanced data in a Decision Tree?
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Happy Learning !!!